



EDS ASSIGNMENT -1

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Subject – EDS





File Handling & Word Counting Program

Introduction

Why do we need File Handling ?

- **Data Volatility**: Standard variables lose data when a program closes. Files give us permanent data storage.
- **Dynamic Inputs**: Instead of hardcoding text into our code, we can read actual text documents, CSVs, and logs.
- **The Universal Workflow**:
 - a. Open the file with specific permissions.
 - b. Process the file (Read or Write).
 - c. Close the file to save memory.

1. File Operations

MODES OF OPENING A FILE

r – open for reading

w – open for writing

a – open for appending

Different modes of Opening a file

SYNTAX

#Opening of file

```
variable = open("File_name" "mode")
```

#Closing of File

```
variable.close()
```

Examples of File Operations

1. Reading a file using Mode - 'r'

```
f = open("File.txt")  
data = f.read()  
print(data)  
f.close()
```

```
f = open("File.txt" , "r")  
data = f.read()  
print(data)  
f.close()
```

Text File(File.txt)

```
File.txt  
1 Hello! I am Atharva Jagtap  
2 I am from Pune  
3 And i study in MITAOE.
```

Output

```
PS D:\Atharva\Python\Code with harry\Chapter 9\EDS Assignment 1>  
ignment 1/Reading.py  
Hello! I am Atharva Jagtap  
I am from Pune  
And i study in MITAOE.
```

Examples of File Operations

2. Writing in a file using Mode - 'w'

```
f = open("File.txt", "w")  
f.write("this is nice")  
f.close()
```

Output

File.txt

```
1 this is nice
```

Text File(File.txt)

File.txt

```
1 Hello! I am Atharva Jagtap  
2 I am from Pune  
3 And i study in MITAOE.
```

Examples of File Operations

3. Appending in a file using Mode - 'a'

```
f = open("File.txt", "a")  
f.write("This is a new text added!")  
f.close()
```

Output

 File.txt

```
1 this is niceThis is a new text added!
```

Text File(File.txt)

 File.txt

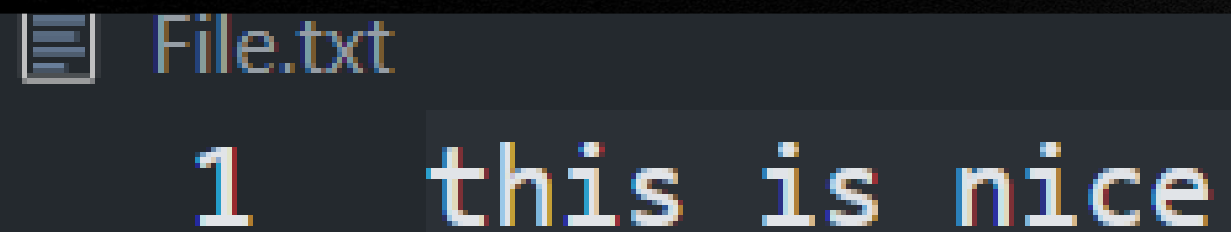
```
1 this is nice
```

Examples of File Operations Using 'with'

Opening File with read mode using 'with open()'

```
with open("File.txt", "r") as f:  
    contents = f.read()  
    print(contents)
```

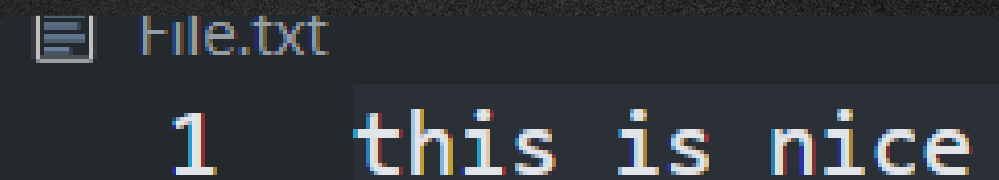
Output



File.txt

```
1  this is nice
```

Text File(File.txt)



File.txt

```
1  this is nice
```

We can use 'with open()' for different basic modes of files!
{ i.e. 'r', 'w' }

The "Character Count" vs. "Word Count" Problem

- **Character Count:** Simply measures the length of the string using `len(content)`. It counts every letter, space, and symbol.
- **Word Count:** Requires logic. We cannot just use `len()`.
- **The Solution:** Python's `.split()` method. It automatically cuts the text wherever there is a space or line break and puts the individual words into a list!

Word Count Problem

```
file = open("File.txt", "r")
text = file.read()
words = text.split()
word_count = len(words)
print("Total words in file:", word_count)
file.close()
```

File.txt

 File.txt

```
1 this is niceThis is a new text added!
```

Output

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Atharva\Python\Code with harry\Chapter 9\EDS Assignment 1>
unt.py"
Total words in file: 8
```

Word Count Problem (Same)

```
with open("File.txt", "r") as f:  
    contents = f.read()  
    print(contents)
```

File.txt

File.txt
1 this is niceThis is a new text added!

Output

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS D:\Atharva\Python\Code with harry\Chapter 9\EDS Assignment 1>
unt.py"
Total words in file: 8

Conclusion & Best Practices

- File operations are foundational to data science, automation, and backend development.
- Always handle potential errors (like a missing file) using Try-Except blocks in real-world applications.



THANKYOU